Volkan Sarp

Google Scholar   theneutron@gmail.com   github/sarpvolkan

Professional Profile

Researcher in solar–terrestrial physics with expertise in observational data analysis and numerical modeling. Experienced in integrating multi-source observations with physics-based simulations to investigate near-Earth space variability. Proven track record in cross-disciplinary methodological transfer. Dedicated to advancing Türkiye's capabilities in ionospheric science and next-generation space weather prediction.

Core Competencies

Space Environment Analysis & Forecasting: Solar cycle variability, active regions, solar flares, ionosphere–thermosphere coupling, traveling atmospheric disturbances; multi-source observational analysis, physics-based simulation, spatiotemporal disturbance characterization, nonlinear time-series analysis, and data-driven forecasting.

Systems Engineering & Collaborative Research: Bridging engineering principles (optimization, systems modeling, process analysis) with advanced space science research. Demonstrated academic contribution with multiple peer-reviewed publications as lead and co-author; active referee for prominent space weather and solar physics journals.

Technical Stack

Programming & Engineering Tools: Python (NumPy, SciPy, Pandas, Xarray, PyWavelets, Matplotlib/Cartopy), MATLAB, R, IDL, Bash/Shell, Fortran/C++ codebase navigation, configuration, and compilation.

Simulation, Modeling & Data Analysis: GITM, IRI-2020, NRLMSIS2.0, FISM2, IGRF; Linux HPC; GNSS-TEC processing (preprocessing, quality control, spatiotemporal mapping), signal processing, nonlinear time-series analysis, and model–data validation.

Research & Engineering Experience

TÜBİTAK-funded Postdoctoral Research Fellow — Jan 2025 – Sep 2025

George Mason University — *Fairfax, VA, USA*

- Developed a multi-scale temporal detrending pipeline for GNSS-TEC observations; characterized scale-dependent ionospheric responses to the April 2024 total solar eclipse.
- Developed a reference-frame transformation for GNSS-TEC analysis that maps observations from a 4D spatio-temporal space onto a co-moving frame that traces the eclipse cone, thereby preserving the coherence of wave structures along the eclipse path.

TÜBİTAK Project Scholar – Fine-Scale Dynamics in Sunspot Umbra — 2017 – 2019

Akdeniz University — *Antalya, Türkiye*

- Adapted and deployed an IDL-based Lagrangian particle-tracking algorithm to automatically detect and track fine-scale solar features across their lifetimes in high-resolution GST/BBSO solar observations.
- Built a statistical analysis workflow for umbral-dot dynamics, enabling one of the largest umbral-dot trajectory analyses reported with GST/BBSO data at the time.

Project Marketing Engineer — Nov 2014 – Jul 2015

Forma Makina Sanayi AŞ — *İstanbul, Türkiye*

- Supported a large-scale ERP implementation program and successfully deployed the CRM system for the sales and marketing department.
- Coordinated customer-facing technical workflows across sales, engineering, and maintenance.

Industrial Engineer – ERP Consultant — Dec 2013 – Oct 2014

Sistaş Sayısal İletişim San. ve Tic. AŞ — *Ankara, Türkiye*

- Supported ERP implementation projects.

Education

Ph.D. in Space Sciences and Technologies — 2018 – 2024

Akdeniz University — *Antalya, Türkiye*

- Thesis: *A Study on the Response of the Thermosphere–Ionosphere System to Solar Flare*
- Configured, executed, and validated GITM general circulation model simulations of an X1.3-class solar flare using FISM2 solar EUV irradiance inputs and flare-off control experiments to isolate radiation-driven ionosphere–thermosphere perturbations.
- Analyzed changes in thermospheric mass density, electron number density, chemical composition, and vertical winds; compared model outputs with observational datasets to assess physical consistency and model fidelity.
- Processed GNSS TEC observations from 2000+ ground stations across North America to characterize multi-scale ionospheric disturbances and identify pronounced longitudinal asymmetries.
- Attributed the observed east–west asymmetry in ionospheric flare response to longitudinal variation in magnetic declination, which modulates ion-drag dissipation of atmospheric gravity waves and thereby structures the background neutral atmosphere differently across meridians.

M.Sc. in Space Sciences and Technologies — 2015 – 2018

Akdeniz University — *Antalya, Türkiye*

- Thesis: *Analysis of Chaotic Features and Modelling of Sunspot Cycle*
- Completed scientific preparation coursework during the transition from engineering to space sciences, including Remote Sensing, Orbital Mechanics, Cameras & Telescopes, and the Sun & Solar System.
- Applied a data-driven nonlinear forecasting approach to solar-cycle variability, using time-delay embedding, state-space reconstruction, and simplex projection to forecast Solar Cycle 25 amplitude and timing.
- Produced an early stronger-than-consensus prediction for the cycle, later supported by its observed evolution.

B.Sc. in Industrial Engineering — 2007 – 2013

TOBB University of Economics and Technology — *Ankara, Türkiye*

- Background in optimization, time-series analysis, systems modeling, CAD/AutoCAD, process analysis, and co-operative industrial training; developed skills in workflow analysis, technical documentation, and structured engineering problem solving.
- Completed a graduation project on electric power transmission planning, applying ARIMA-based time-series forecasting to historical electricity-consumption data for regional demand estimation.

Awards & Fellowships

TÜBİTAK Research Fellowship — George Mason University, USA — **2025**

SCOSTEP Visiting Scholar Award — NASA Goddard Space Flight Center, USA (awarded; deferred) — **2020**

TÜBİTAK Graduate Scholarship — Akdeniz University, Türkiye — **2019 – 2023**

TÜBİTAK Project Scholar — Akdeniz University, Türkiye — **2017 – 2019**

Selected Publications

Sarp, V., Yiğit, E., & Kilcik, A. (2024). Response of the thermosphere-ionosphere system to an X-class solar flare: 30 March 2022 case study. *Space Weather, 22*. doi:10.1029/2024SW003938

Kilcik, A., **Sarp, V.**, Yurchyshyn, V., & Rozelot, J.P. (2020). Physical characteristics of umbral dots derived from high-resolution observations. *Solar Physics, 295*(58). doi:10.1007/s11207-020-01618-y

Sarp, V., Kilcik, A., Yurchyshyn, V., Rozelot, J.P., & Ozguc, A. (2018). Prediction of solar cycle 25: A non-linear approach. *Monthly Notices of the Royal Astronomical Society, 481*(3), 2981–2985. doi:10.1093/mnras/sty2470

10 peer-reviewed journal articles · 2 international conference proceedings — full list: orcid.org/0000-0002-1018-9850